AP BIOLOGY TOPIC 7.6

Evidence of Evolution: Fossils, Anatomy, Biogeography & Molecules

UNIT 7: NATURAL SELECTION (13–20% EXAM WEIGHT)

EVO-1.M | EVO-1.N | HOMOLOGY VS. ANALOGY | MOLECULAR CLOCKS

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 7.6 (EVIDENCE OF EVOLUTION: FOSSILS, ANATOMY, BIOGEOGRAPHY & MOLECULES)

CED Core Vocabulary

Multiple Independent Lines of Evidence — Evolution is supported by converging data from geology, morphology, molecular biochemistry, genetics, biogeography, and mathematical modeling showing that...

Two Methods of Dating Fossils — Relative Dating (Sedimentary Strata): Uses the Law of Superposition — in undisturbed sedimentary rock layers, deeper strata = older and upper strata = younger .

Biogeography — The scientific study of the past and present geographic distribution of species across continents and islands.

Vestigial Structures — Reduced, degenerate, or nonfunctional anatomical remnants inherited from a functional structure in an ancestor. Examples: Pelvic girdle & femur bones inside...

What Fossils Reveal — Document extinct species, chronological succession of life forms, and transitional fossils that bridge major taxonomic groups.

Radiometric (Absolute) Dating — Measures the constant exponential decay of radioactive parent isotopes into stable daughter isotopes using known half-lives ($t_{1/2}$) : Carbon-14 ($^{14}C \rightarrow {}^{14}N$,...

Island Endemics — Oceanic islands (e.g., Galápagos, Hawaii) are colonized by species from the nearest mainland , which then diversify via adaptive radiation.

Comparative Embryology — Early vertebrate embryos share conserved developmental stages — pharyngeal pouches (gill slits) and a post-anal tail — demonstrating shared inherited...

§2. 1. FOSSIL EVIDENCE & GEOLOGICAL DATING

Paleontology

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

1. Fossil Evidence & Geological Dating ➔ **Multiple Independent Lines of Evidence** ➔ **What Fossils Reveal** ➔ **Two Methods of Dating Fossils**

- **Multiple Independent Lines of Evidence:** Evolution is supported by converging data from *geology, morphology, molecular biochemistry, genetics, biogeography, and mathematical modeling* showing that organisms have changed over time and share common ancestry.
- **What Fossils Reveal:** Document extinct species, chronological succession of life forms, and **transitional fossils** that bridge major taxonomic groups.
- **Two Methods of Dating Fossils:**
  - **Relative Dating (Sedimentary Strata):** Uses the *Law of Superposition* — in undisturbed sedimentary rock layers, **deeper strata = older** and **upper strata = younger**.
  - **Radiometric (Absolute) Dating:** Measures the constant exponential decay of radioactive parent isotopes into stable daughter isotopes using known **half-lives ($t_{1/2}$)**:
    - **Carbon-14 ($^{14}C \rightarrow {}^{14}N$, $t_{1/2}$ = 5,730 yr):** Dates young organic remains up to ~50,000–75,000 years old.
    - **Uranium-238 ($^{238}U \rightarrow {}^{206}Pb$, $t_{1/2}$ = 4.5 billion yr) / Potassium-40:** Dates ancient volcanic rock layers bracketing older fossils.

[EXAMPLE & BIOLOGICAL APPLICATION] **Famous Transitional Fossils:**

- ***Tiktaalik roseae* (~375 mya):** Transitional lobe-finned fish → tetrapod with gills/scales AND flat skull, neck, ribs, and wrist-like fin bones.
- ***Archaeopteryx* (~150 mya):** Transitional non-avian dinosaur → bird with flight feathers/wishbone AND teeth, clawed wings, and bony tail.
- ***Pakicetus* / *Ambulocetus*:** Land artiodactyls → modern whales (hind limb reduction).

[CORE AXIOM / REMEMBER] **Rock layers = timeline of Earth history.**

§3. 2. GEOGRAPHICAL EVIDENCE (BIOGEOGRAPHY)

Continental Drift & Islands

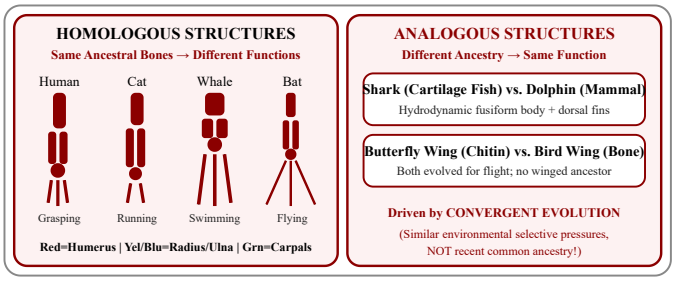
HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

2. Geographical Evidence (Biogeography) ➔ **Biogeography** ➔ **Island Endemics**

- **Biogeography:** The scientific study of the past and present geographic distribution of species across continents and islands.
- **Plate Tectonics & Continental Drift (Pangaea → Gondwana/Laurasia):**
  - Similar fossils (e.g., *Glossopteris* ferns, *Cynognathus* reptiles) and living flightless ratite birds (ostrich, rhea, emu) are found on separated southern continents that were once joined.
  - Geographic isolation via continental breakup reduced gene flow and exposed populations to different selective pressures, driving divergence.
- **Island Endemics:** Oceanic islands (e.g., Galápagos, Hawaii) are colonized by species from the **nearest mainland**, which then diversify via adaptive radiation.

§5. 4. COMPARATIVE ANATOMY: HOMOLOGOUS VS. ANALOGOUS

Divergent vs. Convergent



FEATURE	HOMOLOGOUS STRUCTURES	ANALOGOUS STRUCTURES
Definition	Similar underlying anatomy/development inherited from a **common ancestor** (may have different functions).	Similar superficial function/shape in **unrelated lineages** due to similar environmental pressures.
Evolutionary Pattern	**Divergent Evolution** (e.g., mammalian pentadactyl forelimbs)	**Convergent Evolution** (e.g., insect wing vs. bat wing; shark vs. dolphin)
Phylogenetic Use?	**YES** — used to construct cladograms & infer ancestry.	**NO (Homoplasy)** — can mislead morphological trees!

§6. 5. MOLECULAR HOMOLOGIES: DNA & PROTEIN COMPARISONS

Strongest Evidence

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

5. Molecular Homologies ➔ **DNA, RNA, & Amino Acid Sequence Alignment** ➔ **Highly Conserved Proteins** ➔ **Molecular Clocks**

- **DNA, RNA, & Amino Acid Sequence Alignment:** Provides the most quantitative, objective measure of evolutionary relatedness.
  - **Fewer nucleotide/amino acid differences** = more recent common ancestor.
  - **More sequence differences** = more time has elapsed since lineages diverged.
- **Highly Conserved Proteins:** Essential proteins like **Cytochrome c** (mitochondrial ETC), **Hemoglobin**, and **Histones** are present across diverse species; sequence comparisons map deep evolutionary relationships.
- **Molecular Clocks:** Neutral mutations accumulate at a relatively constant rate over time, allowing scientists to estimate the **time of divergence** between two clades.

[AP EXAM FRQ SCORING STRATEGY] **AP Data Table Skill:** When given a matrix of "Number of Amino Acid Differences in Cytochrome c" (or "% DNA Sequence Identity"), remember: **0–1 differences = closest sister taxa**; largest number of differences = **outgroup** (diverged earliest)!

§4. 3. VESTIGIAL STRUCTURES & COMPARATIVE EMBRYOLOGY

Developmental Homologies

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

3. Vestigial Structures & Comparative Embryology ➔ **Vestigial Structures** ➔ **Comparative Embryology**

- **Vestigial Structures:** Reduced, degenerate, or nonfunctional anatomical remnants inherited from a functional structure in an ancestor.
  - **Examples:** Pelvic girdle & femur bones inside baleen whales and boa constrictors; blind eye remnants in cavefish; human coccyx (tailbone) and appendix.
- **Comparative Embryology:** Early vertebrate embryos share conserved developmental stages — **pharyngeal pouches (gill slits)** and a **post-anal tail** — demonstrating shared inherited developmental gene programs (e.g., *Hox* genes).

§7. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 7.6 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 7.6: Evidence of Evolution: Fossils, Anatomy, Biogeography & Molecules ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 1, 2 & 3 (Artificial Selection, Hardy-Weinberg Mathematical Modeling & BLAST Bioinformatics): Comparing allele frequency shifts across generations under directional selection vs. small-population genetic drift simulations, and constructing maximum-parsimony cladograms from BLAST percentage sequence identity matrices (Cytochrome c / β-globin). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Golden Rule: Hardy-Weinberg Equilibrium (HWE): Allele frequencies $p + q = 1$; Genotype frequencies p^2 (homozygous dominant) + $2pq$ (heterozygous carriers) + q^2 (homozygous recessive) = 1. Golden Calculation Sequence: ALWAYS begin with the recessive phenotype frequency (q^2), take $\sqrt{(q^2)}$ to find q, compute $p = 1 - q$, and then calculate $2pq$ (heterozygote carrier frequency). Never take the square root of the dominant phenotype frequency ($p^2 + 2pq$)! **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Evolutionary & Biomedical Disruption Case Study: Incomplete antibiotic regimens or agricultural antibiotic overuse exert strong directional selection on bacterial populations (e.g., MRSA or MDR-Tuberculosis): susceptible cells die while rare, pre-existing resistant mutants (arising via random mutation or R-plasmid conjugation) survive and reproduce. In conservation biology, severe population bottlenecks (e.g., Northern Elephant Seals, African Cheetahs) drastically erode genetic diversity via genetic drift, increasing homozygosity for deleterious recessive alleles and vulnerability to novel pathogens.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 7.6 to prove why the claim holds.

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